

## CS-1180 Lab Problem 5: Factors

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**PURPOSE:** To test the ability to write nested loops

**DIRECTIONS:** A factor of a number is a value that evenly divides the number, with no remainder. Write a program that prompts the user for a number `n` and prints all of the factors each number from 1 to `n`. If the user enters a value for `n` that is less than 1, display the message `Error: n must be positive`.

For full credit, ensure that your program is well commented to explain your code.

**EXAMPLE:** Spacing, capitalization and wording must match the examples given.

```
What is the value of n?
> 6
1: 1
2: 1 2
3: 1 3
4: 1 2 4
5: 1 5
6: 1 2 3 6

What is the value of n?
> -4
Error: n must be positive
```

### RUBRIC:

#### (3 pts) Base Functionality

- [1] Reads input correctly
- [1] Output is correct
- [1] Handles negative input correctly

#### (1 pts) Style

- [1] The code is meaningfully commented

### IMPORTANT NOTES:

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- Submissions that do not compile will receive a zero
- Submissions with improperly cited AI will receive a zero and an academic integrity violation
- Submissions that are partially or fully copied from another submission will receive a zero and an academic integrity violation